Taxonomy for AI usage declaration in Mathematics

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Abstract

A proposal for how to disclose AI contribution to mathematical research in a standardized way. The proposed disclosure focuses on how much human understanding a paper offers and how much of the work was contributed by AI. For different research tasks, the intensity of AI contribution is measured from 0 to 3. These tasks are `Argument, Intro(duction), Result, Check, Idea, Method, Text, Lit(erature), Code, Data`. Finally, a paper declares its "Human vs. AI understanding taxonomy" class from 0 to 5, which serves as the main identifier for AI contribution in mathematics. Other sciences can use the same labels, but they should propose their own headline identifier that focuses on subject-specific issues.
AI usage classification

```
class3:  Argument1 Intro1 Result3 Idea3 Text1 [Claude Opus 5.5, 2026-10 & Gemini 3.6, 2026-10]
```

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1 Transparacy declaration for AI use in research Mathematics

The goal of this article is to provide a standardized way to declare how much AI, for what purpose and with which intent was used to produce a mathematical research paper. This allows the mathematics community to discuss open and honestly how AI is used. Such discussions should facilitate community accepted ideas for how to deal with the advent of AI in research math. Ideally mathematicians would in fact use the presented standardized discloser for published papers. Anyone who uses it is welcome to provide feedback under `hirschi@math.uni-bonn.de`.

The AI use declaration

Authors submitting mathematical manuscripts provide the following transparency information:

1. Comply with **AIC** in the following section. Hence, provide the AI intensity for the labels `Argument, Intro, Result, Idea, Method, Text, Literature, Code, Data, Check, Translation, AI Referee, OSA, Lean`
2. Determine and declare the overall **AI Autonomy Class** of the paper (`Class` 0–5).
3. Write a **method section** on the use of AI that was not mentioned or sufficiently represented by AIC.
4. Specify which models were used.
5. If a frontier model was used, then declare for which labels the frontier model provided outputs. It would be exemplary to provide the raw input and output for these interactions. However, this is not very practical, and many mathematicians would not like that. Providing the raw data is therefore optional.

Explanation of the term **frontier model**: A frontier model is an LLM of the most recent generation from one of the leading AI labs. For the purposes of this declaration, the term "frontier mode" also encompasses LLMs that are specifically trained to have high performance in logic and mathematics and are comparable in this aspect to other frontier models.

Table 1: Overview of Frontier AI Models (October 2026)

Laboratory	**Model Name**
OpenAI	GPT-6 Astra
Anthropic	Claude Opus 5.5
Google DeepMind	Gemini 4 Argon
DeepSeek	DeepSeek R1

2 AIC, AI Checklist

Analagous to the MSC2020 codes every paper declares in the arXiv abstract the intensity of AI for every research task. Each label below stands for a different research task. For every task a number from 0 to 3 declares the intensity that AI was used to complete this task. The article you are reading would have the following labels
`Argument1 Intro1 Result3 Idea3 Text1 [Claude Opus 5.5, 2026-10 & Gemini 3.6, 2026-10]`
This concept is based on the article [1] "Proposing a Classification Scheme for AI Usage in Scientific Articles" by David Prinz. To support the taxonomy which is the main result of this article I had to change a significant amount of key concpets [1]

Label	Scope
Core Math Labels	
`Argument`	Argument: Justification of the results: proofs, derivations, analysis and interpretation of evidence.
`Intro`	Explanations that do not have proof character. Writing introductions, exposition, proof-sketches, history, known general theory (definitions, statements, proof-adabtions)
`Result`	Formulation of the main results: their statement, sharpening, generalisation or change of assumptions.
`Check`	Verification of computations, code, references, proofs, arguments, explanations.
Meta Labels	
`Idea`	Idea of project: Which research question to work on? Idea for connection with other projects.
`Method`	Approach, design of the study, choice of model, big picture proof strategy.
Supporting Labels	
`Text`	Grammar, orthography, clarity and elegance of prose, typeset (LaTeX), presentation, diagrams, tables, clarity and elegance of formulas and cross-references.
`Lit`	Literature search, summary, citations, bibliography.
`Code`	Software, simulations, symbolic or numerical computations with deterministic algorithms.
`Data`	Collection, cleaning, labelling, plots, synthetic data.
"Binary" Labels	
`Translation`	An AI Translation of a papers in different language was used without human verification.
`AI Referee`	AI refereed the final draft. Disclose the model.
`OSA`	One Shot Attempts: AI was asked to solve the main research question.
`Lean`	A Lean (or comparable) proof was generated and verified. 0 if no Lean proof, 1 Lean proof was written by humans, 2 Lean proof was written largely by AI

Intensity	Name	Meaning
0	None	No AI use (the default; omitted from the tag).
1	Polishing	Minor AI assistance and always under close human supervision. Each task the AI attempted was small, the human always had a very clear vision of the expected output the AI produces.
2	Substantive	AI made a substantive contribution that the authors guided and verified but did not fully determine.
3	Autonomous	AI produced the material essentially autonomously; human input was limited to the initial prompt and final review.

2.1 Further explanations of the labels

`Text` does not only mean the changes AI made or proposed to the text. `Text` encompasses everything in the written presenation: Images, formatting, tables, diagramms, decoration. The idea of `Text` is to combine all the AI contributions that made the content of the paper easier to read and understand as well as nicer to look at. Excluded are however contributions that change the content in any way. Ai contributions to the mathematics be it proofs, explanations or intuitions are not recorded in `Text`.

[1] Scale of intensity from 0 to 3 instead of 0 to 4. The Introduction label is my own idea. Changing the Proof label to the Argument Label. Replacing the headline AI label with the class of the taxonomy. Adding some "Binary" labels. Removing the Fig label. Sharpening and changing the describtion of some labels especially the Check label.

`Text` only changes the presentation and readablilty. `Intro` also can change the content of explanations, introductions, exposition, prof-sketches, history, and known general theory to facilitate a reader understanding the material. AI contributions to `Text` are not allowed to impact the mathematical content while changes to the non-proof-character text sections with mathematical content are recorded in the `Intro` label. The key difference is the task an AI was given. For `Text` the task is to make the text more readable. For `Intro` the task is to explain the mathematical content in a more understandable way.
`Check` has a slightely different scale. `Check` is an interactiv label.

`Intensity`	Name	Meaning
0	None	No AI use to check the content or the form of the paper.
1	Polishing	AI gave useful feedback for the form. The feedback for the content was superficial and did give rise to any substantive changes. The content feedback might be forgotten signs or constant, simple calculations, or forgotten assumptions.
2	Substantive	AI discovered a content issue/error. In fixing the problem, the content / understanding / complexity was significantly changed.
3	Autonomous	AI feedback and AI-proposed changes led to considerable changes that change large parts of the content.

`AI Referee` only applies to the very final draft that is uploaded to a preprint server or submitted to a journal. If there is a retraction or other large-scale changes to the draft, this label asks whether this latest version was again fed in its totality into an LLM and checked for any issues. All previous rounds of feedback from LLM are recorded in the label `Check`.
I strongly suggest disclosing this final AI feedback to the reader, referee, and journal. To avoid conflict, I do not propose to make this disclosure mandatory.

3 Motivation for the Taxonomy

The AIC does not disclose everything that I would like to know about the extent to which AI worked on a paper. Therefore, the very first information is a `Class` from 0 to 5. Why I think this is necessary is explained here. What the classes mean is explained in the next chapter.
I believe that the central goal of mathematics is to provide living human understanding of mathematics, enjoy this understanding, enjoy extending this understanding, actually extend this understanding, and hopefully sometimes produce some real-world applications that are useful to the wider society. Mathematics for me is mostly an art that sometimes has scientific consequences. The two key points are that the understanding needs to be human and living.
`Human`: The understanding is human if a person that understands can explain the content of the knowledge to another person. There are many other ways to describe human understanding, but this version captures most of its meaning.
`Living`: If all the people who understood a topic die because the topic is no longer studied (for example because it is no longer valued) then this knowledge is no longer part of living human understanding. Even if the book and papers still exist, there are no more people able to explain any part of it to someone else. Once mathematicians read the old manuscripts and hopefully understand the written word, the knowledge once again is part of the living human understanding.
I don't think we should produce dead knowledge. Knowledge that is written in a book or in a database but no one understands it well enough to act upon it. Knowledge must be actionable to be truly valuable. This argument founds my belief that we should not value a paper that was first to roughly explain a problem the highest. We should value the best explanation of a problem the highest. Because good explanations will lead to an increase in the longevity of the human understanding.
Therefore, I propose a taxonomy that allows us to talk about the depth of living human understanding of a given paper. Maybe this will someday far in the future help construct value attribution that favors living human understanding. In the meantime I am satisfied if it provides a way to talk about how much AI contributed to the understanding.

The following taxonomy thus is supposed to answer the following two questions:
1) How much living human understanding does a paper offer?
2) How much of the understanding was provided by an AI contribution?
A low class means lots of living human understanding and little AI contribution. A high class means less living human understanding, and lots of AI contributions. For now this is not a value system. I would not claim that a paper of class 4 is less valuable than a paper of class 0. The paper's value is based on its actual contribution. If I need to judge this, then I have to read it and try to understand. The taxonomy aims to describe what the papers offer, not to judge it.
There is one exception. A paper of class 5 is not verified. Noone claims that the content is true. Additionally, no one tried to split it up into manageable pieces that are understood. This contributes nothing at all to the living human understanding yet. Maybe someone else can use the class 5 paper, curate the content, and contribute to the living human understanding. But this achievement would go to the new author that takes words and formulas and produces living human understanding.

4 Taxonomy for Human vs AI understanding in Mathematics

In orange every class specifies the allowed AI contributions by AIC label. Unrestricted means that the intensity of the label is unrestricted (intensity 4 is allowed). I do not give precise boundaries for the restricted labels. The taxonomy is about living human understanding and also the intention of the authors. Until the community has experience assigning the intensity for each label, it is impractical to explain the differences between the classes based on the intensities. As a guideline, I think the restriction is usually capped at intensity 2.

Class 0 | Zero AI

No use of LLMs. Exceptions are strictly limited to spell checkers, code linters (formatting) and searching for literature.

`Text, Lit` restricted

Class 1 | Human-conceived discovery, AI as research assistant

AI was also utilized for literature work, calculations, code, data cleaning and presenation, and write-up (like exposition, introduction, known general theory). AI provided feedback (flagging logic issues without secondary prompting on the nature of the errors or its resolution). This includes discussions with an LLM to refine ideas, claim, and proofs provided that all core mathematical concepts were entirely conceived by a human first, i.e., either by the author or found in the literature.

`Text, Lit, Code, Data, Idea, Intro` unrestricted, `Check, Argument, Result, Method` restricted

Class 2 | AI-prompted discovery with human understanding

At least one significant mathematical idea, claim, or proof (idea) was proposed directly by the AI, without finding an exact reference in the math literature of this idea. The researcher must understand the claim/proof/idea and engage with and explore this idea in a more general context than the proof the AI provided. The researcher develops a deep understanding of the contributions the AI provided. The researcher must generate the precise statement and proof in their own words.

Some parts of the paper were not entirely conceived by a human (or a prior publication of class at most 2) first.

`Check, Result` also unrestricted only `Method, Argument` restricted

Class 3 | Partial AI Generation

Entire claims, chunks of proofs, or whole sections were written directly by an AI without prior conceptual understanding by the human authors. **Even if the human author thoroughly checked, verified, and fully understood the AI's generated work**, the fact that a whole cohesive section of mathematical understanding originated from the AI automatically triggers class 3.

The human is not obligated to engage, understand, and work with the AI output in a deeper way. The authors do not have to rephrase the output in their own words. `Method, Argument` not mostly autonomous

Class 4 | Dominant Autonomous AI Generation

The papers's proofs and core strategy were largely written and structurally conceived by AI. The big picture strategy also originates from the AI.

In class 3, the human researcher choses the intermediate steps of the problem, while in class 4 the progress of understanding was also structured by the AI. `Method, Argument` mostly autonomous. It is verified!

Class 5 | unchecked AI-generated Mathematics

Mathematics generated by an LLM that was not digested, structured or understood by a human. It was neither checked by a human nor by a formal verification tool. This is not allowed to be put on arXiv.

Possible community approach to the issue: The human who curates and understands it can add their own name to the paper and upload as class 4 paper. The generator of the AI slop is not allowed to forbid anyone from curating and claiming the new paper. Not verified!

This classification serves as the headline number to tell the reader, referee or journal how much AI contributed to a mathematical manuscript. It is written in front of all the other labels in the following manner:

`class3: Argument1 Intro1 Result3 Idea3 Text1 [Claude Opus 5.5, 2026-10 & Gemini 3.6, 2026-10]`

Regarding `class` 0: Ideally no LLM would be allowed in class 0. However, it has become increasingly difficult to ensure that digital tools contain no LLMs. A spellchecker might be largely based on the hardcoded grammatical rules but have an LLM integrated that suggests rephrasing to make the wording more elegant. Asking Stack Exchange, Google, or consulting the LaTeXpackage description for how to use certain commands is functionally no different from asking an LLM to explain the errors of LaTeXcode. LLMs may also write LaTeXcode to improve the presentation. Regarding literature work, only one very specific task is allowed in `class` 0. That is searching for literature. (Google) Search is no longer as efficient as it was before the advent of AI. Researchers should not waste their time looking for a reference. This searching is not limitless. Only searching for the documents, not their content, is allowed in class 0. Summaries, automated citations, and work on the bibliography are not permitted in class 0. It is strictly limited to searching for articles.

4.1 Distinguishing the classes

- The mathmatically important AI use cases that are connected to the degree of living human understanding are distributed over the labels `Argument, Intro, Result, Method, Check`. For each label I will explain the tipping point at which I believe that a paper sacrifices some human understanding in order to be complete.

- `Argument` has 2 tipping points.
  First: Any argument that was not first conceived by a human. This can be very small and insignificant. Nevertheless, it detracts from the human discovery process and cedes some of the design power of further exploration to AI. (Difference between C1 and C2)

  Second: If the arguments, their structure, and their integration into known theory are so AI-centric that a human is simply a passenger and no longer takes part in the discovery of understanding. (Difference between C2 and C3)

- `Intro`: When the contextualization and the power of convincing the reader of the truth are dominated by AI-written text, then a human no longer is the source of knowledge but simply the namesake of the content. To convince other people of the correctness of the proof, a human might have to consult the explanations of AI. (Difference between C1, C2, and C3)

- `Resulat`: When AI suggests changes to the assumptions, strength, or generality of a statement that do not originate in human understanding, then the AI dictates the path forward. Small changes that are adjacent to the human proposed ones might not change the dynamic. But if the researcher waits for an upgrade, the AI proposes, then human understanding is no longer the center of concern. (Difference between C1, C2, and C3)

- `Method`: For mathematics that does not involve lots of compute, the label `Method` boils down to the big picture strategy to tackle a problem. If this big picture strategy is proposed by AI, humans are no longer in control of the discovery process. (Difference between C3 and C4)

- `Check`: If a verification attempt falsifies the arguments or statements but suggests changes, this can lead to ideas that were first conceived by AI and not humans. This circumvents human discovery. (Difference between C1 and C2)

This explanation here agrees with the orange labels in the taxonomy above except for the labels `Intro` and `Result`. If a mathematician is outsourcing the proof to the exposition and the introduction, then the label `Intro` could in fact contribute to class 3 status. If a mathematician keeps asking an AI what the minimal assumptions of a statement are and the AI also provides an explanation for why these are the minimal assumptions, then the `Argument` has been outsourced to the search of the framing of the `Result`. In such extreme situations, `Intro` and `Result` can be very influential in the classification. I am hopeful that no one is abusing these labels to this extent.

Much more explanation and edge case discussion will be necessary but here are some important remarks.

- The calculations on Class 1 have to adhere to one of the following concepts.

  1. It is a numerical or quasi-numerical calculation that a calculator/sympy/WolframAlpha/custom program could do. For example, it is allowed to calculate invariants of a concrete example (Gromov-Witten-Invariants of a symplectic manifold) by deterministic rules (product formula or localization formula) with AI and then verify the output yourself.

  2. You expect a certain formula depending on variables. You understand the structure of the expected formula and feel comfortable that you can explain this structure to a human. Then you can ask an AI to determine something depending on variables, e.g. a recursive formula. If it happens to put out something that adheres to your preconceived notion of the output, you stay in Class 1. If you, however, were wrong and the output improved your understanding of the problem, you just used a mathematical idea first conceived by AI, and your paper now has Class 2.

- The feedback from AI to your finished work or to your intermediate results in Class 2 is liberal. You do not need to suppress the explanation of the feedback deliberately. However, if the feedback offers an idea that you cannot attribute to a prior written paper, your own or someone else's human understanding, then you used mathematics first conceived by an LLM, and your paper is now in Class 2.

4.2 Three tiers

The taxonomy is visibly separated into three tiers with a color scheme. This is because classes 0 - 2 represent complete human understanding and a way of approaching mathematics that is uniform among all three classes. Classes 3 - 4 represent AI-conceived discovery that tried to achieve new results quickly while accepting that the contextual understanding of human authors might not be as deep compared to a slower, more in-depth approach of understanding the material. Class 5 does not claim correctness, as it does not require verification and therefore is a separate tier.

- complete human understanding (Almost as before AI)

  AI functions as a writing assistant (formatting, language, search, coding). In dialogue the AI provided small pieces of knowledge, as if one read them in a paper.

  partial human understanding, AI-conceived discovery

  The manuscript is verified by humans or Lean. Some significant part was autonomously generated by AI. Humans only provided minimal oversight at least for thing parts of the manuscript. Proofs are understood by humans, but its consequences were not explored in a broader context.

- no appreciable understanding

  An LLM wrote most of the paper autonomously. The paper is not verified (neither by humans nor by an formal verification system), it is not structured, digested, understood by humans.

4.3 Contagion Rules

I think we need to separate the three tiers of human understanding in the advent of AI. Therefore, I propose a contagion rule with the following purpose:
If a new paper uses the results or techniques of a paper that is based on AI understanding, then the new paper should be treated as if it used AI intensively.
This has two basic justifications.

1. I predict that AI papers will have more and harder to detect mistakes. This increased risk of significant mistakes should be recorded in the metadata of a publication.

2. By citing the content of a paper that was largely written by AI, the chain of human understanding is broken. In practice the author of a new paper will not understand the content of the previous paper, largely originating from AI, better than the human who supervised the AI. Therefore, the new claims are not completely understood by humans.

Obviously there needs to be exceptions and flexibility with this contagion rule. Below I write my proposal with all its exceptions and special cases. I am not convinced that this is a good approach because it is very complicated and not very clear. I expect that the community prefers no contagion rule or a very reduced one. Contamination by a class 5 paper is probably a sensible proposal that almost everybody will agree to. I am therefore very happy to receive feedback on the proposed contagion rules.

- No paper of class 5 is allowed to be referenced without becoming class 5 itself and disappearing into the void of misunderstanding. This only applies if the content of the class 5 paper is used, not if it is merely mentioned.

- If you cite a paper of class 3 or 4, your paper automatically becomes at least class 3 as well.
  Exception: If you completely rewrite the proofs in your own words, verify every intermediate step independently, and adapt or generalize the logic to fit your framework, you can instead publish your work as class 2.
  If the citation is only mentioning other work and not using it, then this rule does not apply.

- A class 5 paper can be changed into a class 2-4 paper by understanding, verifying, rewriting, generalizing, specializing, explaining, presenting, etc., the contents of a class 5 paper. This process needs to be carefully refereed to ensure correctness. I do not have a clear proposal on how to referee this.

- Referencing a class 2 paper does not contaminate your class 0 or 1 status. The reason is that a class 2 paper was completely understood by a human, and the authors can explain all parts of it. Therefore, you are only drawing from the living human understanding of mathematics, which is the same thing mathematicians did before LLMs became capable of outperforming some human mathematical understanding.

- If a paper is class 2 because of a mathematical argument that is not part of the proof of a main result, then the class can be given as `Class-2(n)` where `n` is the number of distinct contributions AI provided to change its class from 1 to 2. For every AI contribution (claim, statement, proof, proof-idea, strategy, or thought-provoking information), there needs to be an explanation. The explanation contains at least the prompt & raw AI output and the reason why it is not a core part of the paper. This special case is designed for long papers with many different statements where very few non-core results contain class-2 AI usage. The paper could have been published without these additional contributions. However, the paper becomes more powerful and useful to the community by including these additional results.

- A class 3 or 4 paper can be changed into a class 2 paper by understanding, verifying, rewriting, generalizing, specializing, explaining, presenting, etc., the contents of a class 3 / 4 paper. This process needs to be carefully refereed to ensure human understanding. I do not have a clear proposal on how to referee this. This should only happen when a researcher is looking for new insights into the material or tries to adapt it to a different situation. A clear plan and reason should be given for attempting this work. This is contrasted by the cleaning up of a class 5 paper, which should be encouraged at any time.
- If the contagion rule forces a paper to become say class 3 but otherwise would have been class 1, then the complete classification is 3C(1). Meaning is of class 3 by contamination, but internally it is of class 1.
  The same works for 5C(4) or any other combination.

5 Acknowledgment

AIC is based on David Prinz's proposal [1]. His original version the AIL2026 will continue as an independent proposal. Originally I planned to make a checklist with all the different tasks an AI might complete during research. David's proposal is, however, much more advanced. To ensure compatibility with my taxonomy, I change a few things. I am very grateful for David's work and input.
I thank David Prinz, Amanda Hirschi, Leo Schelstraete, Naageswaran Manikandan, Sun Woo Park, Félicien Comtat, John Nicholson, Patrick Kennedy Hunt for providing feedback to the taxonomy.

References

[1] Proposing a Classification Scheme for AI Usage in Scientific Articles, David Prinz, 9 October 2026, `https://davidprinz.org/ail2026/`